Computer Games Development CW208
Technical Design Document
Year IV

Olawole Abayomi-Owodunni
C00278711

Olawole Abayomi-Owodunni
C00278711

13/04/2026

Contents

Overview 3
Key Terminology 3
Goals and Technical Requirements 4
Must-Have Goals (Definition of Done) 4
Platform, Tooling and Constraints 4
External Dependencies 4
Software Architecture 5
Architectural Style and Rationale 5
High-Level System Diagram 5
Major Subsystem Overview 6
Ownership and Data Flow 6
Class Design and Specifications 7
Core Class Hierarchy (UML Class Diagram) 7
ICollidable Interface 8
Enemy Hierarchy 8
Steering Behaviours (Strategy Pattern) 8
Projectile Hierarchy 8
Weapon System 9
Room Generation Pipeline 9
CRC Cards 10
CRC Card: Game 10
CRC Card: Player 10
CRC Card: Enemy (abstract) 10
CRC Card: GruntEnemy 10
CRC Card: TurretEnemy 11
CRC Card: FloorGenerator 11
CRC Card: RoomInstance 11
CRC Card: LLMService 11
CRC Card: QuestManager 11

CRC Card: CollisionCheck 11
Algorithm Design 12
Floor Graph Generation Algorithm 12
Floor Layout Embedding Algorithm 12
Corridor Generation Algorithm 13
Collision Detection and Resolution 13
Combat Wave Management 14
LLM Asynchronous Job Queue 15
Runtime Flow 17
State Transitions (Game Modes) 17
Critical Analysis 18
Known Prototype Shortcuts 18
Technical Risks 18
Evaluation Against Goals 19
Conclusions 20
References 21
Books 21
Academic Papers and Reports 21
Web Resources 21

Overview

Key Terminology

Term	Definition
SFML	Simple and Fast Multimedia Library - a cross-platform C++ library providing windowing, graphics, audio, and input functionality.
Roguelike	A subgenre of role-playing games characterised by procedurally generated levels, permadeath, and turn-based or real-time combat.
Hub World	A persistent safe area the player returns to between dungeon runs, used for upgrading and progression.
Layer / Mask Collision Filtering	"Layer" defines what an object is (e.g. PLAYER_LAYER); "Mask" defines what it can collide with. Two objects collide only if each object's layer is present in the other's mask.
ICollidable	A C++ interface (abstract class) that any object participating in collision checks must implement, providing **getCollisionBounds**() and **getCollisionProfile**().
RoomPlan	A lightweight data-only blueprint describing a room's tile layout, doors, spawners, and triggers.
Seeded Procedural Generation	A technique using a deterministic random number generator (std::mt19937) seeded with a known value so that the same seed always produces the same output.
LLM (Large Language Model)	A machine learning model trained on large text dataset, capable of generating human-like text. In this project, the model runs locally via llama.cpp backend and is processed using a custom lightweight C++ wrapper.
Steering Behaviours	Autonomous agent movement algorithms (Seek, Arrive, Flee, etc.) that produce acceleration vectors based on agent state and environment.
AABB	Axis-Aligned Bounding Box - a rectangular collision volume whose sides are parallel to the coordinate axes.

Goals and Technical Requirements

Must-Have Goals (Definition of Done)

The following goals represent the minimum viable product that must be achieved for the project to be considered complete:

- Playable core loop: The player can move, aim, and attack. Enemies spawn and behave consistently. Collisions and damage feel reliable.
- Procedural dungeon structure: Generate Floors and Rooms from a seed. Support core room types (spawn, combat, portal) at minimum.
- Upgrade loop: A hub world exists where currency is earned during runs and the player can spend it on upgrades.
- NPC interaction: NPCs can be interacted with via a chat window. Dialogue is driven by local LLM integration.

Platform, Tooling and Constraints

Requirement	Detail
Language	C++ (C++17 standard)
Graphics / Input	SFML 3.x
IDE / Build	Visual Studio 2026
OS Target	Windows (x64 primary; x86 also tested)
Controller Input	XInput is used to provide controller support
Game Loop	Fixed timestep (~60 FPS); dt passed in seconds
Collision System	Bitmask layer/mask filtering with AABB intersection
Procedural Gen (Seeded)	std::mt19937 seeded with dungeonSeed + floorId for deterministic floor graphs
Procedural Gen (Random)	rand() for per-run room detail variety (obstacle placement, spawns)

External Dependencies

The project relies on the following external dependencies:

- **SFML 3.x:** Windowing, 2D rendering, input handling, and audio.
- **fyp-llm-lib:** A custom C++ wrapper around llama.cpp developed by me, providing **LoadModel**(), **UnloadModel**(), and **Generate**() APIs. Located at external/fyp-llm-lib/llm/LLMWrapper.h.
- **llama.cpp:** The underlying backend C library for LLM inference, used internally by LLMWrapper. Supports GGUF model format files.
- **Llama-3.2-1B-instruct-Q4_K_M.gguf model:** The model being ran to power the game

Software Architecture

Architectural Style and Rationale

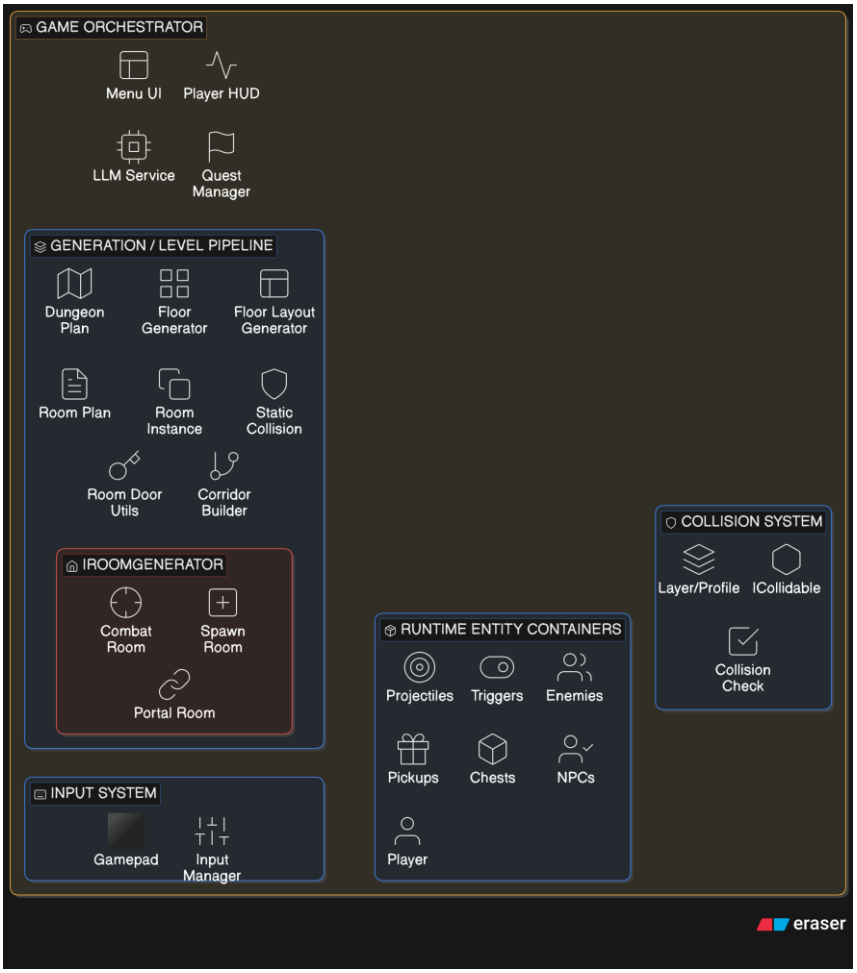
The system follows a centralised orchestrator architecture where the Game class acts as the primary integration point.

Within subsystems, the project uses standard object-oriented patterns:

- Inheritance hierarchies for polymorphic entity types (Enemy: GruntEnemy, TurretEnemy; Projectile: NormalBulletProjectile).
- Interface-based design for collision participants (ICollidable) and room generators (IRoomGenerator).
- Strategy pattern for steering behaviours (ISteeringBehaviour: SeekBehaviour, ArriveBehaviour).
- Service/wrapper pattern for LLM integration (LLMService wraps LLMWrapper with async lifecycle management).
- Data-driven generation using blueprint structs (RoomPlan, FloorPlan, DungeonPlan) that separate data from runtime representation.

High-Level System Diagram

The following text-based diagram illustrates the high-level architecture and data flow between major subsystems (*Generated via eraser.io*):



Major Subsystem Overview

Game Loop: Game manages runtime mode (HUB / DUNGEON), transitions between modes, and acts as the integration point for collisions, room triggers, combat waves, floor transitions, hub interactions and LLM polling.

Player: Player handles movement, aiming, weapon switching, and attacking. It implements ICollidable and contains functions for upgrades, loadout management, and pickup effects.

Enemy / AI: Enemy base class provides shared rendering, health, and collision.
GruntEnemy uses steering (Seek/Arrive) with distance hysteresis to prevent rapidly switching behaviours and melee attacks.
TurretEnemy is stationary with projectile attacks on randomised timers.

Combat / Projectiles / Damage Triggers: Weapons gate fire by cooldown and emit either Projectiles (ranged) or DamageTriggers (melee). Game owns runtime containers and handles lifecycle, collision, and cleanup.

Collision: Layer/mask bitmask filtering with AABB intersection. ICollidable interface is implemented by all collidable entities. Response is centralised in **Game::CollisionChecks**().

Room / Floor / Dungeon Generation: Three-tier procedural pipeline: DungeonPlan => FloorPlan (graph) => RoomPlans (blueprints) => RoomInstances (runtime). Floor graphs are deterministic (seeded) whilst Room details use **rand**() for variety.

LLM Integration: LLMService wraps LLMWrapper with async init/request/poll lifecycle. Game adds a job queue layer (LLMJobContext) for room/NPC/quest prompts, submitting one job at a time when idle.

UI / HUD: PlayerHUD renders health/ammo/coin/quest overlay. MenuUI manages screen flow (main menu, pause, gameplay, game over, quest board)

Quest System: QuestManager maintains a 3-slot quest board with kill and fetch quest types. Progress is tracked during dungeon runs and rewards are committed on return to hub.

Hub Economy / Upgrades: Game owns currency and upgrade state. Shops support weapon, ammo and player stat upgrades, cosmetic cycling, and armory weapon swapping.

Ownership and Data Flow

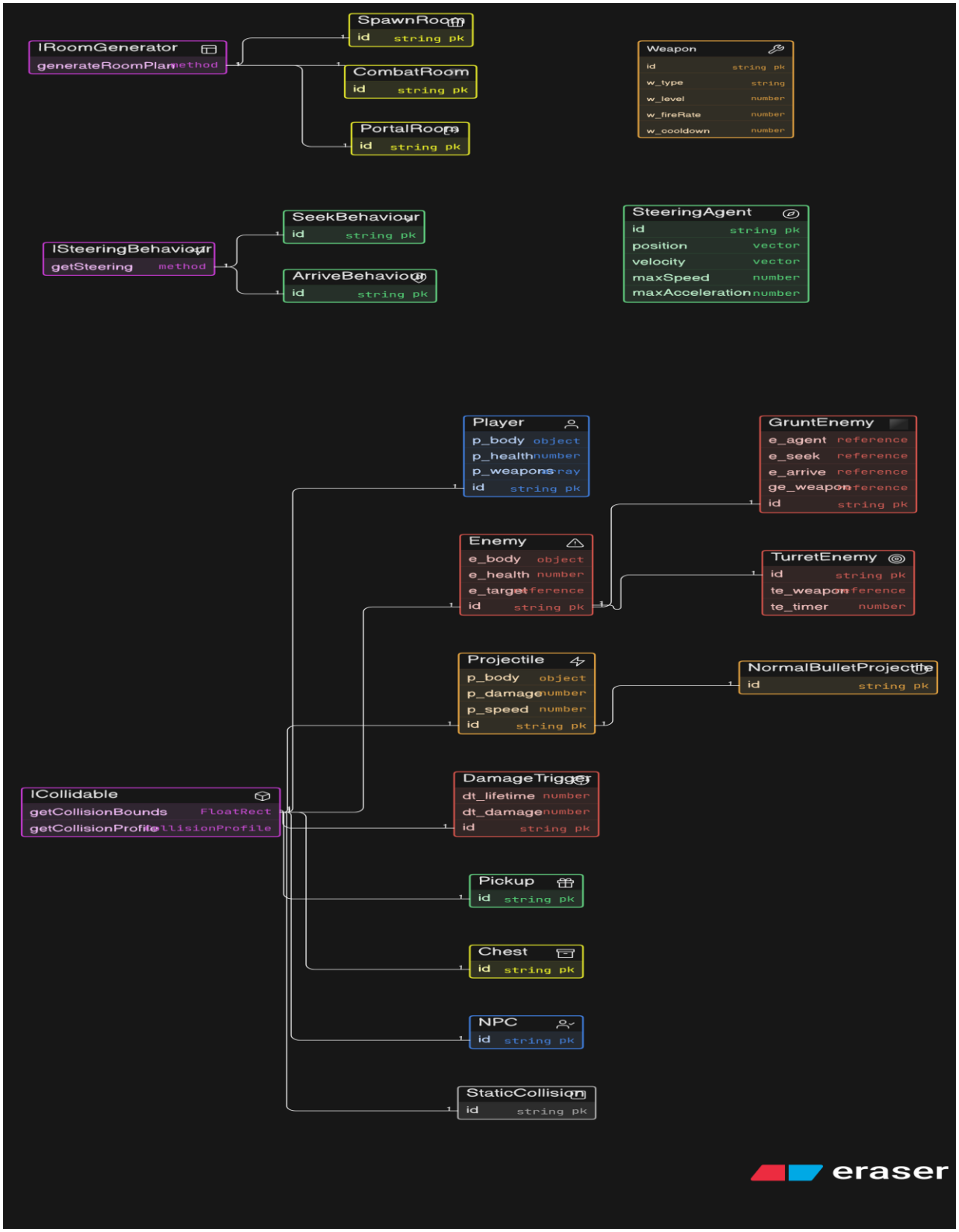
The Game class is the sole owner of all major runtime state. This centralised ownership model simplifies lifecycle management and makes data flow explicit:

- Player (single instance)
- PlayerHUD, MenuUI (single instances)
- QuestManager (single instance)
- Enemies: std::vector<std::unique_ptr<Enemy>>
- Projectiles: std::vector<std::unique_ptr<Projectile>>
- Damage triggers: std::vector<std::unique_ptr<DamageTrigger>>
- Pickups: std::vector<std::unique_ptr<Pickup>>
- Chests: std::vector<std::unique_ptr<Chest>>
- Hub NPCs: std::vector<NPC>

Class Design and Specifications

Core Class Hierarchy (UML Class Diagram)

The UML class diagram captures some of the primary inheritance and composition relationships in the system (*Generated via eraser.io*):



ICollidable Interface

ICollidable is the central interface for the collision system. Any object that participates in collision checks must implement this interface. It requires two pure virtual methods:

- **getCollisionBounds**() => sf::FloatRect: Returns the axis-aligned bounding box of the object.
- **getCollisionProfile**() => CollisionProfile: Returns the object's collision layer and mask.

The CollisionProfile struct contains a CollisionLayer (*bitmask flag identifying what the object is*) and a uint32_t mask (*identifying what layers the object can collide with*). The **canCollideWith**() method performs bidirectional mask checking: A.mask & B.layer AND B.mask & A.layer must not return 0.

Enemy Hierarchy

The Enemy class is an abstract base providing shared body setup/rendering, target tracking, collision profile, health/death state, and damage handling. Derived classes must implement **init**(), **update**(float dt), **setTarget**(), **hitWall**(), and **getEnemyType**().

<u>GruntEnemy:</u> A steering-driven melee attacker. It owns a SteeringAgent and switches between SeekBehaviour (when far from target) and ArriveBehaviour (when close) using distance hysteresis thresholds. Within arrive range, it uses an EnemyWeapon (KNIFE) to spawn DamageTrigger instances for melee attacks.

<u>TurretEnemy:</u> A stationary ranged attacker. It tracks the player's direction and fires projectiles using an EnemyWeapon (PISTOL) on a randomised cooldown timer. It does not move but implements hitWall() for consistency with the base class contract.

Steering Behaviours (Strategy Pattern)

The steering system follows Reynolds' (1999) model and implements the Strategy design pattern. ISteeringBehaviour is an abstract interface with a single pure virtual method: **getSteering**(const SteeringAgent& agent) => Vector2f.

SteeringAgent is a lightweight struct holding position, velocity, maxSpeed, and maxAcceleration. The agent is updated each frame by applying the steering output as acceleration.

<u>SeekBehaviour</u>: Calculates a desired velocity directly toward the target at maximum speed, then returns the steering vector (desired - current velocity), clamped to maxAcceleration.

<u>ArriveBehaviour</u>: Similar to Seek but with a deceleration zone. When the agent is within the slowRadius, the desired speed is scaled proportionally to distance, producing a smooth deceleration to the target.

Projectile Hierarchy

Projectile is an abstract base class providing body rendering, lifetime tracking, collision support (via ICollidable), damage output, and destruction signalling. The collision profile depends on the source: player-fired bullets use PLAYER_BULLET_LAYER (masking ENEMY_LAYER,), while enemy-fired bullets use ENEMY_BULLET_LAYER (masking PLAYER_LAYER) with both profiles masking (WALL_LAYER, DOOR_LAYER).

NormalBulletProjectile: The current concrete bullet type. It normalises direction on init, moves by direction × speed × dt each frame, and expires by either lifetime expiration or collision destruction.

Weapon System

The Weapon class supports both ranged and melee weapons. It gates fire by cooldown and delegates projectile/trigger creation to pure virtual methods (spawnProjectile, spawnMeleeHit). WeaponType enum consists of a weapons set and each weapon type configures its own fire rate, damage, fire mode, and rumble intensity.

Player weapons use a loadout model with a maximum of 3 equipped slots. Weapons are rebuilt from loadout entries via **buildWeaponsFromLoadout**() in the player class. Weapon level scaling adjusts damage and fire rate tuning. Ranged attacks consume ammo from the player's shared ammo pool, with shotgun consuming more ammo per attack than other ranged weapons.

Room Generation Pipeline

The room generation pipeline separates data from runtime representation using a two-stage workflow:

Stage 1 (Blueprint Generation): IRoomGenerator implementations (CombatRoom, SpawnRoom, PortalRoom) produce RoomPlan structs. A RoomPlan contains metadata (id, type, seed), dimensions (width, height, tileSize), a 1D tile map (indexed as row × width + column), and vectors of DoorPlan, SpawnerPlan, and TriggerPlan entries. CombatRoom randomises width (15–20) and height (10–13), places outer walls, random interior obstacles, and spawner tiles proportional to interior area. SpawnRoom and PortalRoom use fixed sizes (11×11).

Stage 2 (Runtime Build): RoomInstance.**buildFromPlan**() converts a RoomPlan into renderable SFML shapes and StaticCollision colliders by going through each tile in the 1D array and creating a physical tile based on the tile type of each tile. Wall tiles become wall shapes + WALL_LAYER colliders. Door tiles become door shapes with optional DOOR_LAYER colliders (when locked). Trigger plans become colliders with appropriate layer flags (PORTAL_TRIGGER_LAYER, DOOR_TRIGGER_LAYER).

CRC Cards

The following Class-Responsibility-Collaborator (CRC) cards summarise the primary responsibilities and collaborations for key classes in the system:

CRC Card: Game

Responsibilities	Collaborators
• Own SFML window and drive game loop • Manage game mode (HUB/DUNGEON) and transitions • Update and render all subsystems • Centralise collision checks • Manage combat waves and floor transitions • Route menu actions and hub interactions • Poll LLM service and dispatch results • Manage economy/upgrade state	Player Enemy Projectile DamageTrigger Pickup Chest NPC MenuUI PlayerHUD QuestManager LLMService RoomInstance FloorGenerator DungeonPlan

CRC Card: Player

Responsibilities	Collaborators
• Handle movement input (keyboard/gamepad) • Handle aiming input (mouse/right stick) • Manage weapon loadout and switching • Emit projectiles and damage triggers • Track health/ammo state • Respond to wall collisions • Apply upgrades and pickup effects	InputManager Weapon Projectile DamageTrigger ICollidable

CRC Card: Enemy (abstract)

Responsibilities	Collaborators
• Provide shared body setup and rendering • Track target position • Manage health/death state • Implement ICollidable interface	ICollidable CollisionProfile

CRC Card: GruntEnemy

Responsibilities	Collaborators
• Steer toward player using Seek/Arrive • Switch behaviours by distance hysteresis • Attack with melee weapon (KNIFE)	SteeringAgent SeekBehaviour ArriveBehaviour EnemyWeapon DamageTrigger

CRC Card: TurretEnemy

Responsibilities	Collaborators
• Track player direction • Fire projectiles on randomised timer	EnemyWeapon Projectile

CRC Card: FloorGenerator

Responsibilities	Collaborators
• Generate deterministic floor graph from seed • Enforce room type constraints (spawn first, portal last) • Prevent direct spawn-portal edges	FloorPlan FloorNode FloorEdge std::mt19937

CRC Card: RoomInstance

Responsibilities	Collaborators
• Build SFML shapes from RoomPlan • Build StaticCollision colliders for walls/doors/triggers • Render room geometry	RoomPlan StaticCollision

CRC Card: LLMService

Responsibilities	Collaborators
• Manage async model init/shutdown • Queue generation requests on worker thread • Poll for completed responses • Report readiness/busy state	LLMWrapper std::thread std::mutex std::atomic

CRC Card: QuestManager

Responsibilities	Collaborators
• Maintain 3-slot quest board • Generate kill and fetch quests • Track active quest progress • Finalise run results and commit rewards	QuestData EnemyType PickupType

CRC Card: CollisionCheck

Responsibilities	Collaborators
• Perform AABB intersection with layer/mask filtering	ICollidable CollisionProfile

Algorithm Design

Floor Graph Generation Algorithm

The FloorGenerator produces a deterministic floor graph (FloorPlan) from a seed. The algorithm ensures that the floor is always solvable and follows structural constraints.

```
FUNCTION generateFloorPlan(floorId, dungeonSeed, isBossFloor):
    floorSeed = dungeonSeed + floorId
    rng = mt19937(floorSeed)

    roomCount = randomInt(rng, 3, 10)

    // Create room nodes
    rooms[0] = {id: 0, type: SPAWN}
    FOR i = 1 TO roomCount - 2:
        rooms[i] = {id: i, type: COMBAT}
    rooms[roomCount-1] = {id: roomCount-1, type: PORTAL}

    // Generate edges (connections)
    edges = []
    FOR i = 1 TO roomCount - 1:
        // Connect each room to a random earlier room
        parent = randomInt(rng, 0, i - 1)

        // Prevent direct spawn → portal edge
        IF parent == 0 AND i == roomCount - 1:
            IF roomCount > 2:
                parent = randomInt(rng, 1, i - 1)

        edges.add({id_a: parent, id_b: i})

    // Build adjacency list
    plan.buildAdjacencyList()

    RETURN plan
```

Analysis: Time complexity is O(n) where n is the room count. The use of std::mt19937 ensures deterministic output for a given seed, enabling reproducible floor layouts for testing and seeded runs.

Floor Layout Embedding Algorithm

FloorLayoutGenerator embeds the floor graph into a 2D grid, producing world-space positions for each room. The algorithm uses a BFS traversal to place rooms without overlaps.

```
FUNCTION generateFloorLayout(floorPlan, roomPlans):
    grid = empty 2D occupied-cells map
    positions = array of Vector2f[roomCount]

    // Place spawn room at origin
    positions[0] = (0, 0)
    markOccupied(grid, room[0], (0, 0))

    queue = BFS queue starting from room 0

    WHILE queue is not empty:
        currentId = queue.dequeue()

        FOR EACH neighbour IN adjacencyList[currentId]:
            IF neighbour is already placed:
                CONTINUE
```

```
            // Try directions: EAST, SOUTH, WEST, NORTH
            FOR EACH direction IN {E, S, W, N}:
                candidatePos = positions[currentId] + offset(direction, roomSize)

                IF NOT overlaps(grid, room[neighbour], candidatePos):
                    positions[neighbour] = candidatePos
                    markOccupied(grid, room[neighbour], candidatePos)
                    queue.enqueue(neighbour)
                    BREAK

    RETURN positions
```

Analysis: The BFS traversal ensures that rooms are placed in proximity order from the spawn room, producing compact layouts. The overlap check prevents room collisions. In the worst case, if all preferred directions are occupied, the algorithm may need to search further, but for typical floor sizes (3–10 rooms) this is not a practical concern.

Corridor Generation Algorithm

Corridors are generated to physically connect rooms that share a floor graph edge. Each corridor is built as a temporary RoomPlan of type CORRIDOR and immediately converted to a RoomInstance.

```
FUNCTION buildCorridors(floorPlan, roomPlans, roomWorldPositions):
    FOR EACH edge IN floorPlan.edges:
        roomA = roomPlans[edge.id_a]
        roomB = roomPlans[edge.id_b]

        // Find matching door pairs (one door on each side)
        doorA = findDoorFacing(roomA, directionOf(roomA → roomB))
        doorB = findDoorFacing(roomB, directionOf(roomB → roomA))

        // Calculate world-space door positions
        startPos = worldPosition(roomA) + tileToWorld(doorA)
        endPos   = worldPosition(roomB) + tileToWorld(doorB)

        // Build corridor plan: a narrow strip of floor tiles bordered by walls on
both sides
        corridorPlan = buildCorridorStrip(startPos, endPos, tileSize)

        // Convert to runtime instance
        corridorInstance = new RoomInstance()
        corridorInstance.buildFromPlan(corridorPlan, corridorWorldPos)

        appendToRoomInstances(corridorInstance)
```

Collision Detection and Resolution

The collision system uses a two-phase approach: layer/mask filtering followed by AABB intersection. This is evaluated for all relevant entity pairs each frame in Game::CollisionChecks().

```
FUNCTION areColliding(entityA: ICollidable, entityB: ICollidable) → bool:
    profileA = entityA.getCollisionProfile()
    profileB = entityB.getCollisionProfile()

    // Phase 1: Layer/mask filtering
    IF (profileA.mask & profileB.layer) == 0:
        RETURN false
```

```
    IF (profileB.mask & profileA.layer) == 0:
        RETURN false

    // Phase 2: AABB intersection
    boundsA = entityA.getCollisionBounds()
    boundsB = entityB.getCollisionBounds()

    RETURN boundsA.intersects(boundsB)
```

```
FUNCTION CollisionChecks():
    // Player vs enemies
    FOR EACH enemy IN enemies:
        IF areColliding(player, enemy):
            player.takeDamage(contactDamage)

    // Projectiles vs enemies/player
    FOR EACH projectile IN projectiles:
        FOR EACH enemy IN enemies:
            IF areColliding(projectile, enemy):
                enemy.takeDamage(projectile.applyDamage())
                projectile.destroy()
        IF areColliding(projectile, player):
            player.takeDamage(projectile.applyDamage())
            projectile.destroy()

    // Entities vs walls/doors (rollback response)
    FOR EACH wall IN staticColliders:
        IF areColliding(player, wall):
            player.hitWall()  // restore previous position
        FOR EACH enemy IN enemies:
            IF areColliding(enemy, wall):
                enemy.hitWall()
        FOR EACH projectile IN projectiles:
            IF areColliding(projectile, wall):
                projectile.destroy()

    // Pickups/Chests vs player
    FOR EACH pickup IN pickups:
        IF areColliding(player, pickup):
            applyPickupEffect(player, pickup)
            pickup.destroy()

    // Portal/Door triggers
    FOR EACH trigger IN triggers:
        IF areColliding(player, trigger):
            handleTriggerInteraction(trigger)
```

Analysis: The current approach is $O(n^2)$ for pairwise checks. This is acceptable for the prototype's typical entity counts (< 50 entities per room). If entity counts scale significantly, a broadphase spatial partitioning structure (e.g., spatial hash grid or quadtree) would be required to maintain performance.

Combat Wave Management

Combat rooms support a wave-based spawning system. When the player enters an uncleared combat room via a door trigger, combat begins:

```
FUNCTION onDoorTrigger(roomId):
    IF roomPlans[roomId].type == COMBAT AND NOT roomPlans[roomId].isCleared:
        startCombat(roomId)
```

```
FUNCTION startCombat(roomId):
    m_isInCombat = true
```

```
    m_waveCounter = totalWaves  // typically 2-3 based on room size
    lockDoors(roomId)
    generateNewWave(roomPlans[roomId])
    spawnEnemies(roomId)
```

```
FUNCTION ManageWave():
    IF NOT m_isInCombat:
        RETURN

    IF enemies.isEmpty():
        m_waveCounter = m_waveCounter - 1

        IF m_waveCounter <= 0:
            // Room cleared
            m_isInCombat = false
            unlockDoors(m_activeRoomId)
            roomPlans[m_activeRoomId].isCleared = true
        ELSE:
            // Next wave
            generateNewWave(roomPlans[m_activeRoomId])
            spawnEnemies(m_activeRoomId)
```

LLM Asynchronous Job Queue

The LLM integration uses an asynchronous architecture to prevent blocking the game thread. The LLMService manages worker threads for both model initialisation and text generation. The Game class adds a queue layer to serialise multiple LLM requests.

```
// LLMService async lifecycle
FUNCTION initAsync(modelPath):
    IF already initialising: RETURN false
    llm_initInProgress = true
    SPAWN thread: initWorkerMain(modelPath)
    RETURN true
```

```
FUNCTION initWorkerMain(modelPath):
    success = llm_wrapper.LoadModel(modelPath)
    LOCK(llm_initMutex):
        llm_initResult = success
        llm_isReady = success
    llm_initInProgress = false
```

```
FUNCTION requestGenerate(prompt):
    IF NOT ready OR busy: RETURN false
    llm_isBusy = true
    SPAWN thread: generateWorkerMain(prompt)
    RETURN true
```

```
FUNCTION generateWorkerMain(prompt):
    response = llm_wrapper.Generate(prompt)
    LOCK(llm_resultMutex):
        llm_latestResponse = response
    llm_isBusy = false
```

```
// Game job queue layer
FUNCTION processLLMQueue():
    IF llmService.isBusy() OR llmJobQueue.empty():
        RETURN

    job = llmJobQueue.front()
    llmJobQueue.pop()
    m_currentLLMJob = job
```

```
    m_hasActiveLLMJob = true
    llmService.requestGenerate(job.prompt)
```

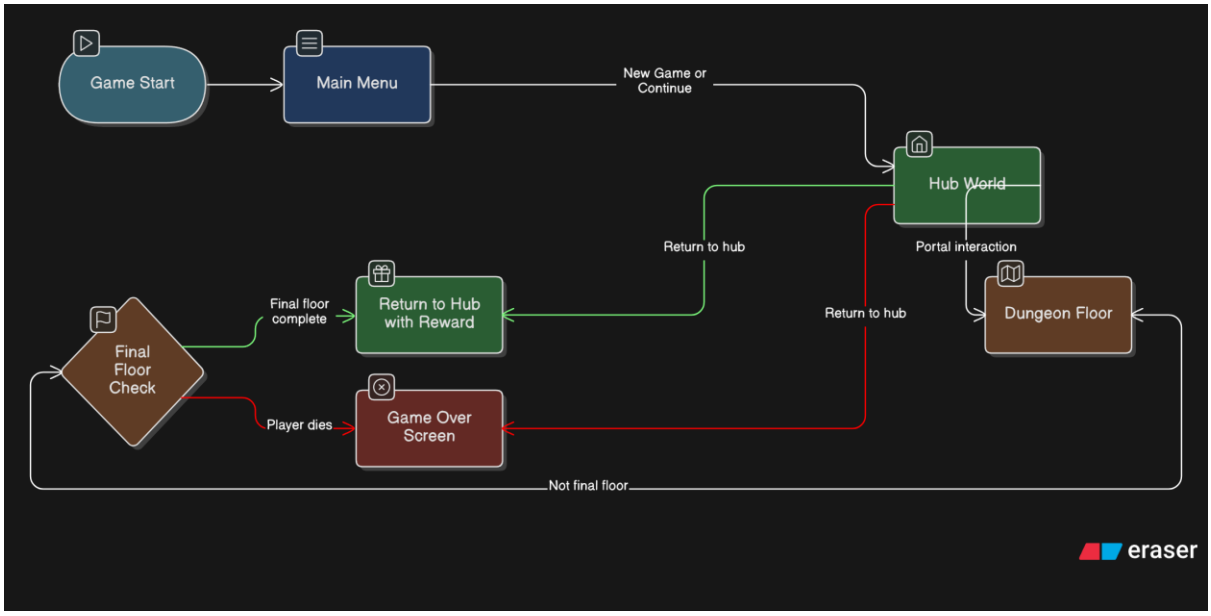
```
FUNCTION update(dt):
    // ... other updates ...

    // Poll init completion
    initResult = llmService.tryConsumeInitResult()
    IF initResult.has_value():
        IF initResult == true:
            queueQuestMetadataJobs()

    // Poll generation completion
    response = llmService.tryConsumeLatestResponse()
    IF response.has_value():
        handleLLMResponse(response.value())
        m_hasActiveLLMJob = false

    processLLMQueue()
```

Analysis: This architecture ensures the game thread never blocks on LLM operations. The queue serialises requests to prevent concurrent model access. Thread safety is achieved through std::atomic flags and std::mutex-protected shared data. The tryConsume pattern (optional return) allows non-blocking polling.

Runtime Flow

State Transitions (Game Modes)

The game operates in two primary modes with defined transitions (*Generated by eraser.io*):



Critical Analysis

Known Prototype Shortcuts

The following prototype shortcuts have been identified and documented. Each represents a conscious trade-off between development velocity and production readiness:

Collision response: Uses simple position rollback (restore previous position on collision) rather than sliding or impulse-based resolution. This can cause "sticking" against walls when moving diagonally. A sliding response would improve player feel.

Collision broadphase: All collision checks are brute-force pairwise comparisons ($O(n^2)$). Acceptable for current entity counts (<50 per room) but would require spatial partitioning (spatial hash grid or quadtree) for larger scenes.

Non-deterministic room details: Room obstacle/spawner placement uses rand() rather than seeded generators. While acceptable for gameplay variety, this means replay/deterministic testing of individual rooms is not currently possible.

Game class complexity: The Game class acts as a "god object" for orchestration. While this is a common and pragmatic approach for prototypes, it accepts many responsibilities and would benefit from extraction of subsystems (e.g., CollisionManager, CombatManager, HubManager) for easier expansion.

Corridor generation: Procedural corridor building may need additional edge-case handling for alignment and overlap situations.

LLM response handling: Text is consumed as complete responses (no streaming). Response validation is lightweight (simple tagged-line parsing for quest text). No formal latency budget policy exists.

Technical Risks

Risk	Severity	Mitigation
Performance degradation	Medium	If entity counts scale beyond current levels without broadphase collision, frame rate will degrade. Mitigation: implement spatial hash grid when entity counts exceed ~100 per room.
Complexity growth	Medium	As room types, interactions, and combat systems expand, the centralised Game class will become increasingly difficult to maintain. Mitigation: extract subsystem managers progressively.
LLM latency	Low–Med	LLM generation takes ~5.5s in Release x64 and ~30s in Debug. Long generation times could create dead spots in gameplay. Mitigation: pre-queue prompts during loading; implement caching for repeated contexts; consider streaming output.
LLM integration stability	Low	Moving from debug triggers to production dialogue flow requires robust error handling and fallback templates. Mitigation: implement response validation with fallback defaults for malformed LLM output.
Corridor alignment	Low	Edge cases in corridor generation could produce misaligned or overlapping corridor geometry. Mitigation: add alignment constraints and overlap detection in corridor builder.

Evaluation Against Goals

This section evaluates the current prototype against the must-have goals defined in Goals section:

Goal	Status	Evidence / Notes
Playable core loop	Achieved	Player can move (WASD/gamepad), aim (mouse/right stick), and attack with multiple weapon types. Enemies spawn with consistent AI behaviours (Seek/Arrive for Grunts, ranged for Turrets). Collision and damage are functional with layer/mask filtering.
Procedural dungeon	Achieved	Floors are generated from seeds with 3–10 rooms per floor. Three room types (Spawn, Combat, Portal) are implemented. Multi-floor progression with portal triggers is functional.
Upgrade loop	Achieved	Hub world exists with shop zones for weapon upgrades, player stat upgrades, ammo upgrades, cosmetic cycling, and armory weapon swapping. Currency (coins) is earned during runs from enemy kills and pickups.
NPC interaction	Achieved	Hub NPCs can be interacted with via overlap + input. LLM integration provides async dialogue generation. Quest board NPCs are linked to the quest system.
Quest system	Achieved	QuestManager supports kill and fetch quest types with a 3-slot board. Progress tracking and reward payout on run completion are functional.
LLM integration	Achieved	Async model loading, room description generation, NPC reply generation, and quest metadata generation are all functional.

Conclusions

This Technical Design Document has presented the complete technical architecture, class design, algorithm specifications, and critical analysis of my game.

The project demonstrates several significant technical achievements:

- A working procedural dungeon generation pipeline that produces playable, interconnected floor layouts from a single seed value, using graph-based generation with BFS embedding.
- A modular collision system with bitmask layer/mask filtering, supporting diverse entity types (players, enemies, projectiles, damage triggers, pickups, chests, NPCs, walls, doors, and triggers) through a shared ICollidable interface.
- An AI system using Reynolds' steering behaviours (Seek, Arrive) with a clean Strategy pattern implementation, enabling enemy movement that is both effective and extensible.
- A complete weapon system supporting six weapon archetypes across ranged and melee categories, with a 3-slot loadout model, level scaling, and distinct fire modes.
- A novel locally-hosted LLM integration for dynamic content generation (room descriptions, NPC dialogue, quest lore), implemented with an asynchronous job queue architecture that prevents blocking the game thread.
- A hub-and-dungeon meta-progression loop with economy, upgrades, quest board, and NPC interactions.

The critical analysis has identified several areas where the prototype could be improved:

- Collision response could be enhanced from simple rollback to sliding resolution for smoother player movement against walls.
- The Game class's centralised orchestration role should be progressively decomposed into dedicated subsystem managers as feature scope grows.
- Broadphase spatial partitioning should be implemented before entity counts scale significantly beyond current levels.
- LLM response validation should be strengthened with formal schema checking and fallback templates for malformed output.
- Room generation could be made fully deterministic by replacing rand() calls with per-room seeded generators.

The design decisions documented throughout this TDD reflect a prototype-first philosophy: build simple, working systems first, validate gameplay rules, then refactor toward production-quality architecture.

Future work would focus on implementing the stretch goals (additional room types, procedural boss generation, save/load persistence), hardening the LLM integration for production dialogue flows, and progressively decoupling the centralised Game class into dedicated subsystem managers.

References

Books

[1] Gregory, J. (2014). Game Engine Architecture. 2nd ed. Boca Raton: CRC Press.

[2] Millington, I. and Funge, J. (2009). Artificial Intelligence for Games. 2nd ed. Burlington: Morgan Kaufmann.

[3] Nystrom, R. (2014). Game Programming Patterns. Genever Benning.

[4] Ericson, C. (2004). Real-Time Collision Detection. San Francisco: Morgan Kaufmann.

Academic Papers and Reports

[5] Reynolds, C. W. (1999). Steering Behaviors For Autonomous Characters. Proceedings of Game Developers Conference 1999, San Jose, CA. Miller Freeman Game Group.

Web Resources

[6] SFML Team. (2024). SFML - Simple and Fast Multimedia Library. [Online]. (https://www.sfml-dev.org/). (Accessed 13 April 2026).

[7] Gerganov, G. (2024). llama.cpp - Inference of Meta's LLaMA model in pure C/C++. [Online]. (https://github.com/ggerganov/llama.cpp). (Accessed 13 April 2026).

[8] Microsoft. (2024). XInput Game Controller APIs. [Online]. (https://learn.microsoft.com/en-us/windows/win32/xinput/). (Accessed 13 April 2026).

[9] cppreference.com. (2024). std::mt19937 - Mersenne Twister pseudo-random generator. [Online]. (https://en.cppreference.com/w/cpp/numeric/random/mersenne_twister_engine). (Accessed 13 April 2026).

[10] eraser.io. (2025) AI for Technical Design and Documentation diagram generator. [Online]. (https://www.eraser.io/). (Accessed 13 April 2026).